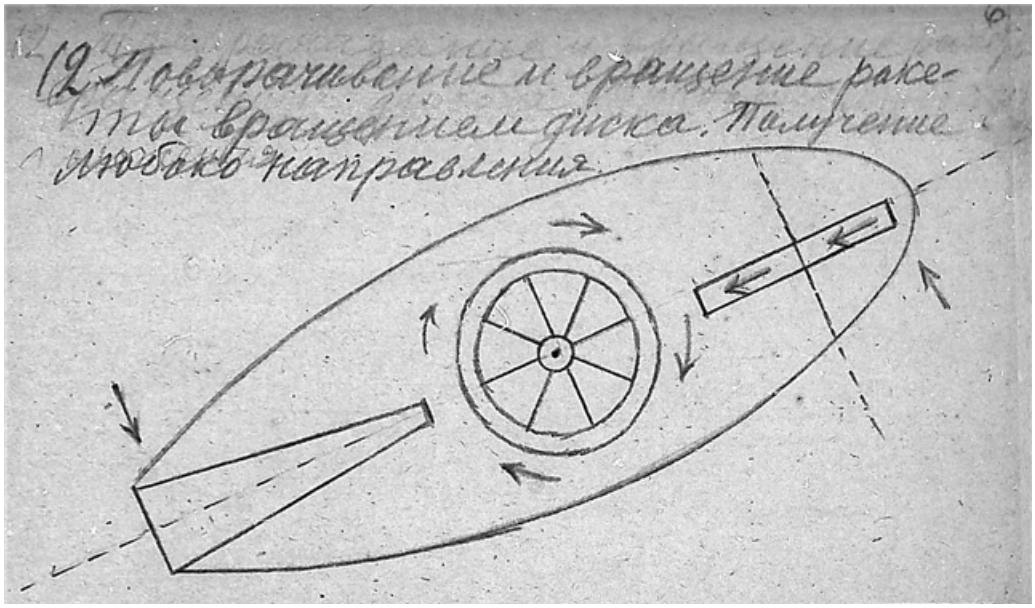
0. Write down the names of any people you worked with and any resources (online or otherwise) that you used.

Computing spacecraft rotations



In HW 4, we considered rotating by 90° using the middle wheel, followed by rotating 90° using the right wheel. You can reference the posted solutions. As before, we will let Rotation 1 denote the 90° rotation using the middle reaction wheel from HW 4 Question 4, and we will let Rotation 2 denote the 90° using the right reaction wheel from HW 4 Question 5. Also, as before, we will let Rotation 3 denote the rotation that results from applying Rotation 1 followed by Rotation 2.

In HW 4 Question 6, we understood Rotation 3 directly, without using quaternions. In this homework, we'll use quaternions. Let r_1 and r_2 denote the quaternions for Rotations 1 and 2 that you found in HW 4 Questions 4 and 5. Let r_3 denote the quaternion for Rotation 3. We will find r_3.

1. (a) Find r_3 using quaternion multiplication. See HW 3 Question 10. Be careful with the order.

   Warning: the rule $e^{x+y} = e^x e^y$ works if x and y are real numbers, and it works if x and y are complex numbers, but it *does not work* if x and y are quaternions. In general, this rule works if $xy = yx$, which is always true for real or complex numbers, but usually not true for quaternions.

   Solution: Based on HW 3 Question 10, the resulting quaternion is

$$r_3 = r_2 r_1 = \left(\tfrac{\sqrt{2}}{2} - \mathbf{k}\,\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{\sqrt{2}}{2} + \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right) = \left(\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{\sqrt{2}}{2}\right)(1-\mathbf{k})(1+\mathbf{j})$$
$$= \tfrac{1}{2}(1 + \mathbf{j} - \mathbf{k} + \mathbf{i}) = \tfrac{1}{2} + \tfrac{1}{2}\,\mathbf{i} + \tfrac{1}{2}\,\mathbf{j} - \tfrac{1}{2}\,\mathbf{k}.$$

   (b) Based on the quaternion you found in the previous part, what is the axis and angle of this rotation?

Hint: Don't forget that $e^{\mathbf{u}\theta} = \cos\theta + \mathbf{u}\sin\theta$ represents a rotation by 2θ, not θ.

Solution: The real part of the quaternion is $\cos\theta$, so $\cos\theta = \frac{1}{2}$, and so $\theta = 60°$. Thus, our result is a rotation by $2\theta = 120°$. Meanwhile, $\sin\theta = \frac{\sqrt{3}}{2}$, so we can compute the axis of rotation $\mathbf{u}$ using

$$\begin{aligned} \mathbf{u}\left(\tfrac{\sqrt{3}}{2}\right) &= \tfrac{1}{2}\mathbf{i} + \tfrac{1}{2}\mathbf{j} - \tfrac{1}{2}\mathbf{k}, \\ \mathbf{u} &= \tfrac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} - \mathbf{k}). \end{aligned}$$

Note that the problem did not require us to express the axis as a unit vector, so $\mathbf{i}+\mathbf{j}-\mathbf{k}$ is also an acceptable answer.

Note also that $\cos\theta = \frac{1}{2}$ also has a solution $\theta = -60°$, in which case $\sin\theta = -\frac{\sqrt{3}}{2}$. This is also an acceptable solution; you would obtain a rotation by $2\theta = -120°$ about $\mathbf{u} = \frac{1}{\sqrt{3}}(-\mathbf{i} - \mathbf{j} + \mathbf{k})$. All that's going on here is that a counterclockwise rotation about one axis is the same as a clockwise rotation about the opposite axis.

2. What happens if we apply Rotation 1 and Rotation 2 in the other order? For the purposes of this question, Rotation 3 is now the rotation you get by applying Rotation 2 followed by Rotation 1.

   Using either the direct method from HW 4 or the quaternion method from the previous question, determine the axis and angle of Rotation 3. For this question, you are only required to do one method, though of course you can check your work by doing both.

Solution: Since the steps are pretty much the same as above, we're just going to do a quick summary.

This time, we have $r_3 = r_1 r_2$, so

$$r_3 = \left(\tfrac{\sqrt{2}}{2} + \mathbf{j}\,\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{\sqrt{2}}{2} - \mathbf{k}\,\tfrac{\sqrt{2}}{2}\right) = \tfrac{1}{2}(1 + \mathbf{j})(1 - \mathbf{k}) = \tfrac{1}{2}(1 - \mathbf{i} + \mathbf{j} - \mathbf{k}).$$

Following the same reasoning as before, this is a 120° rotation, but its axis is now different; it is

$$\mathbf{u} = \tfrac{1}{\sqrt{3}}(-\mathbf{i} + \mathbf{j} - \mathbf{k}).$$

If we compare this axis to the one we found before, we see that $\mathbf{i}$ became $-\mathbf{i}$. It's a different axis of rotation.

Following the other method, we make the chart.

Applying Rotation 2 to the positive x-axis, we obtain the negative y-axis. Applying Rotation 1 to the negative y-axis, we obtain the negative y-axis.

Applying Rotation 2 to the positive y-axis, we obtain the positive x-axis. Applying Rotation 1 to the positive x-axis, we obtain the negative z-axis.

Applying Rotation 3 to the positive z-axis, we obtain the positive z-axis. Applying Rotation 1 to the positive z-axis, we obtain the positive x-axis.

In summary,

$$\text{Rotation 3:}\quad \begin{array}{ccc} \text{positive } x\text{-axis} & \mapsto & \text{negative } y\text{-axis} \\ \text{positive } y\text{-axis} & \mapsto & \text{negative } z\text{-axis} \\ \text{positive } z\text{-axis} & \mapsto & \text{positive } x\text{-axis} \end{array}$$

Applying this rotation three times, we compute

$$\begin{array}{ccccccc} \text{positive } x\text{-axis} & \mapsto & \text{negative } y\text{-axis} & \mapsto & \text{positive } z\text{-axis} & \mapsto & \text{positive } x\text{-axis} \\ \text{positive } y\text{-axis} & \mapsto & \text{negative } z\text{-axis} & \mapsto & \text{negative } x\text{-axis} & \mapsto & \text{positive } y\text{-axis} \\ \text{positive } z\text{-axis} & \mapsto & \text{positive } x\text{-axis} & \mapsto & \text{negative } y\text{-axis} & \mapsto & \text{positive } z\text{-axis} \end{array}$$

As before, since applying this rotation three times got us back to where we started, it is a $120°$ rotation.

To find the axis, we write the chart as

$$\text{Rotation 3:}\quad \begin{array}{ccc} \langle a,0,0\rangle & \mapsto & \langle 0,-a,0\rangle \\ \langle 0,b,0\rangle & \mapsto & \langle 0,0,-b\rangle \\ \langle 0,0,c\rangle & \mapsto & \langle c,0,0\rangle \end{array}$$

Adding, we obtain

$$\text{Rotation 3:}\quad \langle a,b,c\rangle \quad \mapsto \quad \langle c,-a,-b\rangle$$

If we are on the axis, then $a = c$, $b = -a$, and $c = -b$, which, in short, is $a = -b = c$. Thus, any multiple of $\langle 1,-1,1\rangle$ is on the axis. In particular, this matches our answer using quaternions, since $\mathbf{u} = -\frac{1}{\sqrt{3}}\langle 1,-1,1\rangle$.

Ditching the third wheel

In the figure at the beginning of this homework, there are only two reaction wheels, which allow rotation about the y and z axes (using the same system of coordinates we've been using before). There is no third wheel that allows rotation about the x-axis. However, we saw in HW 4 Question 2 that we can still get a $180°$ rotation about the x-axis indirectly, by using the two wheels we have one after the other. But what if we want a rotation about the x-axis by a different angle, like $26°$ or whatever? In this homework, we will discuss a couple options for solving this problem.

3. Let Rotation 1 be a $90°$ rotation about the y-axis. Let Rotation 2 be a 2θ rotation about the z-axis. Let Rotation 3 be the composition of the following three rotations, in this order:

   1. Apply Rotation 1 in reverse.
   2. Apply Rotation 2.
   3. Apply Rotation 1.

   Let r_1, r_2, and r_3 be the quaternions corresponding to Rotations 1, 2, and 3, respectively. So that we're all on the same page about signs, let $r_1 = e^{\mathbf{j}\frac{\pi}{4}}$ and let $r_2 = e^{\mathbf{k}\theta}$.

   (a) Explain why $r_3 = r_1 r_2 \overline{r}_1$ using facts we've already learned.

Solution: As we saw in HW 2 Question 11, the quaternion corresponding to Rotation 1 in reverse is $\overline{r}_1$. As we saw in HW 3 Question 10, to compose rotations, we multiply the corresponding quaternions in the reverse order. Thus, the above composition of rotations is given by the quaternion $r_1 r_2 \overline{r}_1$, as desired.

(b) Compute r_3.

Warning: Same warning as before: For quaternions, $e^q e^r$ is usually *not* equal to e^{q+r}. That exponent rule only works if qr is equal to rq, which is usually not the case for quaternions (but is always the case for real numbers and complex numbers). If you were to try to use this "rule" here, you'd end up with $e^{\mathbf{j}\frac{\pi}{4}+\mathbf{k}\theta-\mathbf{j}\frac{\pi}{4}} = e^{\mathbf{k}\theta}$, which is just r_2, and incorrect.

Solution: We must compute

$$r_3 = \left(\tfrac{\sqrt{2}}{2} + \mathbf{j}\tfrac{\sqrt{2}}{2}\right)(\cos\theta + \mathbf{k}\sin\theta)\left(\tfrac{\sqrt{2}}{2} - \mathbf{j}\tfrac{\sqrt{2}}{2}\right).$$

There are many ways to do this computation, but a particularly nice way is to factor out the $\frac{\sqrt{2}}{2}$ terms and split up the computation according to $\cos\theta$ or $\sin\theta$. We compute

$$\begin{aligned}
r_3 &= \tfrac{1}{2}\left((1+\mathbf{j})\cos\theta(1-\mathbf{j}) + (1+\mathbf{j})(\mathbf{k}\sin\theta)(1-\mathbf{j})\right) \\
&= \tfrac{1}{2}(\cos\theta(1+\mathbf{j})(1-\mathbf{j}) + \sin\theta(1+\mathbf{j})\mathbf{k}(1-\mathbf{j}) \\
&= \tfrac{1}{2}\left(\cos\theta(1-\mathbf{j}+\mathbf{j}+1) + \sin\theta(\mathbf{k}-\mathbf{kj}+\mathbf{jk}-\mathbf{jkj})\right) \\
&= \tfrac{1}{2}(2\cos\theta + \sin\theta(\mathbf{k}+\mathbf{i}+\mathbf{i}-\mathbf{k})) \\
&= \tfrac{1}{2}(2\cos\theta + 2\mathbf{i}\sin\theta) \\
&= \cos\theta + \mathbf{i}\sin\theta.
\end{aligned}$$

(c) Confirm that r_3 is a 2θ rotation about the x-axis.

Solution: A rotation by 2θ about the x-axis is given by

$$e^{\mathbf{i}\theta} = \cos\theta + \mathbf{i}\sin\theta,$$

which matches what we found in the previous part.

We'll now generalize the result from the previous problem.

4. Let Rotation 1 be a rotation by the angle 2φ about the axis $\mathbf{u}$. Let Rotation 2 be a rotation by the angle 2θ about the axis $\mathbf{v}$. As before, let Rotation 3 be the composition of the following three rotations, in this order.

   1. Apply Rotation 1 in reverse.
   2. Apply Rotation 2.
   3. Apply Rotation 1.

   Let $\mathbf{w}$ be the vector that $\mathbf{v}$ gets sent to by Rotation 1. Show that Rotation 3 is a rotation by 2θ about the axis $\mathbf{w}$.

In other words, we have learned how to rotate the axis of rotation. The operation that inputs r_2 and outputs $r_1 r_2 \overline{r}_1$ yields a rotation that is the same as r_2 except that its axis has been rotated according to r_1.

Hint: Compute r_3 as in the previous problem. You will need to write out r_2 in terms of sines and cosines. However, you can save yourself some effort by *not* writing out r_1 in terms of sines and cosines; just leave it as r_1 and use what you've learned.

Solution: As suggested, we compute

$$\begin{aligned} r_3 &= r_1(\cos\theta + \mathbf{v}\sin\theta)\overline{r}_1 \\ &= r_1\overline{r}_1\cos\theta + r_1\mathbf{v}\overline{r}_1\sin\theta. \end{aligned}$$

Since r_1 is a unit quaternion, we know that $r_1\overline{r}_1 = |r_1|^2 = 1$. (Or we can compute explicitly using $r_1 = \cos\varphi + \mathbf{u}\sin\varphi$.) Meanwhile, by the quaternion rotation formula, $r_1\mathbf{v}\overline{r}_1$ is the vector $\mathbf{v}$ gets sent to by Rotation 1, namely $\mathbf{w}$. Therefore,

$$r_3 = \cos\theta + \mathbf{w}\sin\theta.$$

This is precisely the quaternion corresponding to a rotation by 2θ about the axis $\mathbf{w}$, as desired.

In the next two questions, we'll get back to the problem of rotating about the x-axis using only two wheels on the y and z axes. One of the issues with the previous approach is that Rotation 1 was a large 90° rotation. Let's say you have a communications satellite, trying to keep its beam pointed towards Earth. Most of the time, you only need to make small changes to the spacecraft orientation to keep the beam pointed the right way. The beam might start to drift away from where it needs to be a little, and then you fix it. But if fixing it requires doing a large 90° rotation, your beam will be pointing completely the wrong way until you're done, making your satellite useless for several minutes. That's no good. Perhaps we can solve this problem using only small rotations?

Let ε be a small number, and let Rotation 1 be a rotation by an angle of 2ε about the y-axis, and let Rotation 2 be a rotation by an angle of 2ε about the z-axis. Let Rotation 3 be the composition of the following four rotations, in this order:

1. Apply Rotation 2 in reverse.
2. Apply Rotation 1 in reverse.
3. Apply Rotation 2.
4. Apply Rotation 1.

In the next two questions, we will show that Rotation 3 is approximately a rotation by an angle of $4\varepsilon^2$ about the x-axis.

5. First of all, what does "approximately" mean here? In this problem, it means that $\varepsilon^3 \approx 0$. The intuition is that if ε is 0.01, then $\varepsilon^2 = 0.0001$, and $\varepsilon^3 = 0.000001$. In this problem, we've decided that 0.0001 is still big enough to be significant, but 0.000001 is small enough that we can ignore it. (Why did we decide that ε^2 is significant but not ε^3? If we had said that ε^2 is insignificant, we would find that the rotation angle of Rotation 3 is approximately zero,

which is correct but uninformative, so no good. On the other hand, if we had said that ε^3 is significant, we would have to do more work, which is also no good.)

Since $\varepsilon^3 \approx 0$, we also have $\varepsilon^4 = \varepsilon^3\varepsilon \approx 0\varepsilon = 0$, and likewise $\varepsilon^5 \approx 0$, and so forth. Using Taylor series, we therefore have

$$\sin\varepsilon = \varepsilon - \frac{\varepsilon^3}{6} + \frac{\varepsilon^5}{5!} \pm \cdots \approx \varepsilon,$$
$$\cos\varepsilon = 1 - \frac{\varepsilon^2}{2} + \frac{\varepsilon^4}{4!} \pm \cdots \approx 1 - \frac{\varepsilon^2}{2}.$$

Using these approximations along with regular algebra, crossing out all terms with ε^3, ε^4, and so forth, approximate the following quantities.

(a) $\cos^2\varepsilon$.

Solution: We have

$$\cos^2\varepsilon \approx \left(1 - \frac{\varepsilon^2}{2}\right)^2 = 1 - \varepsilon^2 + \frac{\varepsilon^4}{4} \approx 1 - \varepsilon^2.$$

(b) $\cos^4\varepsilon$. Hint: Use the previous part.

Solution: We have

$$\cos^4\varepsilon = \left(\cos^2\varepsilon\right)^2 \approx \left(1 - \varepsilon^2\right)^2 = 1 - 2\varepsilon^2 + \varepsilon^4 \approx 1 - 2\varepsilon^2.$$

(c) $\cos^2\varepsilon\sin^2\varepsilon$.

Solution: We have

$$\cos^2\varepsilon\sin^2\varepsilon \approx (1 - \varepsilon^2)\varepsilon^2 = \varepsilon^2 - \varepsilon^4 \approx \varepsilon^2.$$

(d) $\sin(\varepsilon^2)$.

Solution: Plugging in ε^2 for ε in the Taylor series for $\sin\varepsilon$, we find

$$\sin\varepsilon^2 \approx \varepsilon^2.$$

(e) $\cos(\varepsilon^2)$.

Solution: Plugging in ε^2 for ε in the Taylor series for $\cos\varepsilon$, we find

$$\cos\varepsilon^2 \approx 1 - \frac{\varepsilon^4}{2} \approx 1.$$

6. Now, compute (approximately) the quaternion corresponding to Rotation 3 and verify that it is approximately a rotation by an angle of $4\varepsilon^2$ about the x-axis. To keep us on the same page about signs, let $r_1 = e^{\mathbf{j}\,\varepsilon}$ and $r_2 = e^{\mathbf{k}\,\varepsilon}$.

   Hint: I recommend the following approach. Writing r_1 and r_2 in terms of cosine and sine, the expression for r_3 is the product of four binomials, which gives 16 terms. Compute these terms as follows. First compute the $\cos^4\varepsilon$ terms (one term). Then compute the $\cos^3\varepsilon\sin\varepsilon$ terms (four terms). Then compute the $\cos^2\varepsilon\sin^2\varepsilon$ terms (six terms, be very very careful about signs and order). Then compute the $\cos\varepsilon\sin^3\varepsilon$ terms ... but wait! Isn't $\sin^3\varepsilon \approx 0$? Justify this, and then don't compute these terms. Likewise for the $\sin^4\varepsilon$ terms. Simplify what's left, and then use the previous part to approximate. Then write out the quaternion corresponding a rotation by $4\varepsilon^2$ about the x-axis, and again use the previous part to approximate. You should hopefully obtain the same answer.

Solution: We have that

$$\begin{aligned}
r_3 &= r_1 r_2 \overline{r}_1 \overline{r}_2 \\
&= (\cos\varepsilon + \mathbf{j}\sin\varepsilon)(\cos\varepsilon + \mathbf{k}\sin\varepsilon)(\cos\varepsilon - \mathbf{j}\sin\varepsilon)(\cos\varepsilon - \mathbf{k}\sin\varepsilon) \\
&= \cos^4\varepsilon + (\mathbf{j} + \mathbf{k} - \mathbf{j} - \mathbf{k})\cos^3\varepsilon\sin\varepsilon + (\mathbf{jk} - \mathbf{jj} - \mathbf{jk} - \mathbf{kj} - \mathbf{kk} + \mathbf{jk})\cos^2\varepsilon\sin^2\varepsilon \\
&\qquad + (\cdots)\cos\varepsilon\sin^3\varepsilon + (\cdots)\sin^4\varepsilon \\
&\approx \cos^4\varepsilon + (\mathbf{i} - 1 - \mathbf{i} + \mathbf{i} + 1 + \mathbf{i})\cos^2\varepsilon\sin^2\varepsilon \\
&\approx 1 + 2\mathbf{i}\varepsilon^2.
\end{aligned}$$

Here, we used $\sin^3\varepsilon \approx \varepsilon^3 \approx 0$, so $\cos\varepsilon\sin^3\varepsilon \approx 0$ and $\sin^4\varepsilon \approx 0$.

On the other hand, we can consider a rotation by an angle of $4\varepsilon^2$ about the x-axis. We have $2\theta = 4\varepsilon^2$, so $\theta = 2\varepsilon^2$ and so this rotation is given by the quaternion

$$\cos(2\varepsilon^2) + \mathbf{i}\sin(2\varepsilon^2) \approx 1 + \mathbf{i}(2\varepsilon^2).$$

Here we approximated similarly to parts (d) and (e) above. We observe that we obtained the same answer, so Rotation 3 is indeed approximately a rotation by an angle of $4\varepsilon^2$ about the x-axis.

Note that, in practice, it's easier to just send up a spacecraft with three wheels. But if one of them breaks unexpectedly, you can keep these ideas in mind.

That's all for quaternions. We'll move on to differential forms next with a brief interlude about launch windows.